



MOTHERBOARD

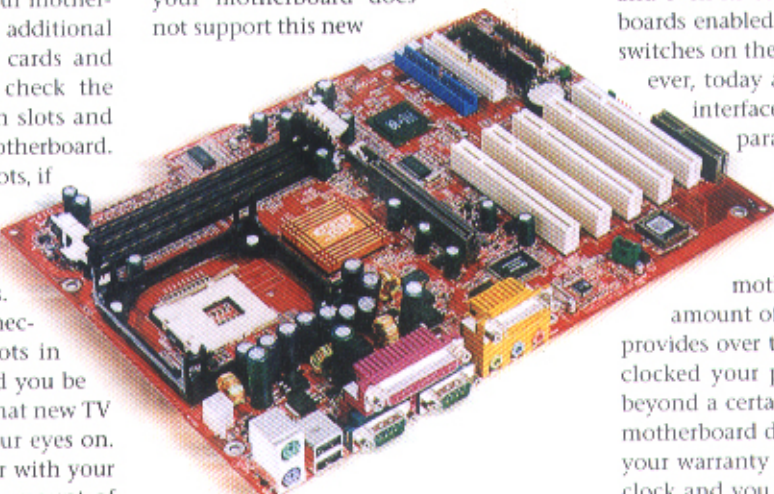
The motherboard has numerous points of control, and these offer many pathways for sapping the horsepower out of your computer

This miniature city of components and interconnections forms the physical, as well as the electronic infrastructure on which the rest of your system is based. Being such a complex component, there are numerous settings, tweaks and considerations that can alleviate any throttle points in your system.

If you are the kind who loves to pack your system with the latest components, you have to make sure that your motherboard is capable of supporting additional components such as add-on cards and memory sticks. You should check the number of free PCI expansion slots and RAM slots available on your motherboard. When it comes to expansion slots, if you want speed and compatibility, it is advisable to stay away from motherboards with the older ISA slots. Make sure that you have the necessary number of free PCI slots in your system. Only then would you be able to buy components like that new TV tuner card that you've got your eyes on. Another very important factor with your system's overall speed is the amount of RAM in your system. With just a simple RAM upgrade, you could garner anywhere between 20 and 40 per cent increase in your system's performance. Added RAM gives your applications more room to exist in and minimises the need to page to your hard disk—a definite bottleneck.

One important consideration while upgrading your processor is whether it will be compatible with your motherboard. The latest processors incorporate technologies and capabilities which require specific support from your motherboard. With both Intel and AMD having moved over to the 0.13-micron technology, their newer processors have different voltage and power requirements compared to older predecessors. This requires special support from the motherboard. Therefore, if you need a faster processor, ensure that your motherboard either supports it implicitly or that there is a BIOS upgrade available for it on the manufacturer's Web site.

There is a wide range of interfaces on motherboards that decide which types of internal and external components you can add to your computer. Your CD-ROM, DVD and hard disk drives are all connected to your motherboard's IDE channels. If you've just bought a new 7,200-rpm ATA/133 hard disk drive and haven't noticed any perceptible performance increases, it could mean that your motherboard does not support this new



This city of components can either boost or buck your system's performance

IDE technology. Even though your hard disk will work fine on an older interface, you will not see a performance increase unless your motherboard also supports the new technology. Another classic case is when you buy a brand new 'USB only' printer and find out that you cannot connect it to your computer, as the motherboard doesn't have the necessary ports. Mice, keyboards and gaming peripherals are based on a number of interfaces—make sure the one you plan to use is supported by the motherboard.

Gaming enthusiasts who are constantly on the lookout for a graphics card upgrade will face a definite roadblock if their integrated graphics motherboard does not have an AGP slot; the only way out is to upgrade the motherboard to one that does. Also, make sure that the

motherboard supports an AGP standard such as AGP 4x or higher. This is the speed that most of today's graphics cards are based upon. Upcoming motherboards will be based on the AGP 8x standard but this is still some time away from becoming mainstream.

Every motherboard offers methods by which you can set parameters such as your processor's clock speed, its bus speed and even its core voltage. Older motherboards enabled this through tiny thumb switches on the motherboard itself. However, today all motherboards offer an interface in the BIOS where these parameters can be modified.

An easy way to experience a speed up is to overclock your CPU.

The overclockability of a motherboard is dictated by the amount of control the motherboard provides over these features. If you overclocked your processor but couldn't go beyond a certain speed, it's because your motherboard does not allow it. However, your warranty becomes void if you overclock and you need to be careful so that you do not damage your processor or motherboard.

Tips & Tweaks

- Load the latest chipset drivers from the company Web site to ensure peak motherboard performance and to resolve known issues with your board's chipset.
- Enable performance settings from your motherboard's BIOS: set System BIOS to cacheable and enable 'Passive Release and Dynamic Bursting' for the PCI bus.
- Use tools such as Motherboard Monitor or the utilities bundled with your board that let you monitor aspects of your system such as processor temperature and fan speeds. This helps trap potential bottlenecks before they escalate into more serious problems.